Name:	**Version A, KEY**
Student ID Number:	

Managerial Economics (ARE) 106
University of California, Davis, Summer Session I, 2024
Dr. John H. Constantine

Midterm

July 10, 2025

Test is from: 10:00 a.m. – 11:00 a.m.

60 TOTAL POINTS

4 PAGES

Show all work: No work, no credit.

Problem 1 (24 points):
Given are the OLS results for estimating the factors in determining the price of a home. The dependent variable is PRICE (or, y = price). The right-hand-side variables (the x's) for the house are the square footage (sqft), the age (age), whether the house is located in a desirable neighborhood (utown), if there is a pool (pool), and if there is a fireplace (fplace).

price	sqft	age	utown	pool	fplace
205.452	23.46	6	0	0	1
185.328	20.03	5	0	0	1
248.422	27.77	6	0	0	0
154.69	20.17	1	0	0	0
221.801	26.45	0	0	0	1
199.119	21.56	6	0	0	1
272.134	29.91	9	0	0	1
250.631	27.98	0	0	0	1
197.24	24.8	0	0	1	0
235.755	27.5	0	0	0	0
189.639	20.82	14	0	0	0

Regression Statistics	
Multiple R	0.9320
R Square	0.8686
Adjusted R Square	0.8679
Standard Error	15.3344
Observations	1,000

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	5	1,544,711.83	308,942.37	1,313.84	0
Residual	994	233,734.31	235.15		
Total	999	1,778,446.14			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	6.91	4.29	1.61	0.11	-1.51	15.33
sqft	**A**	0.17	49.76	0.00	7.99	8.65
age	-0.19	0.05	-3.74	0.00	-0.29	-0.09
utown	60.20	0.97	**B**	0.00	58.29	62.10
pool	4.35	1.21	3.61	0.00	1.99	6.72
fplace	1.40	**C**	1.43	0.15	-0.52	3.32

For any problems that ask you to conduct a hypothesis test, you are to state the test you are conducting and state your results. $\alpha = 0.05$.

(a) What is the value of one additional square foot? Is the estimate statistically significant? (6 points).

This is box A on the printout. Using the t-test equation, you find that $b_1 = 8.46$. A sqarefoot adds, on average, \$8.46.
H_0: $\beta_2 = 0$, H_A: $\beta_2 \neq 0$. t-hat =49.76; t – critical = 1.960. Reject H_0. The estimate is statistically significant.

(b) If a house is located in utown, does this, on average, add value to the house, and if so, how much? You are to calculate t-critical as well. Conduct a two-tailed hypotheses test (assume $\alpha = 0.05$). (6 points)

This is box B. t-hat = 62.06; t – critical = 1.960. Reject H_0. On average, a house located in utown will add \$60.20.

(c) Suppose we expect $\beta_6 > 0$. Conduct a one-tail test to determine if having a fireplace adds value to the house. (6 points)

H_0: $\beta_2 \leq 0$, H_A: $\beta_2 > 0$. t-hat = 1.43; t – critical = 1.645. Do not reject H_0. A fireplace doesn't statistically significantly add value to the house.

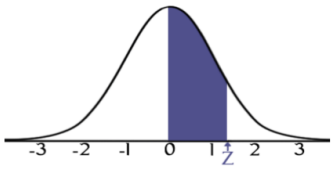
(d) Conduct the test that, simultaneously, none of the right-hand-side variables combined do not affect the price of a home. (6 points)

H_0: $\beta_2 = \beta_3 = \ldots = \beta_6 = 0$
H_A: not H_0.

F-hat = 1,313.84

F_C ($\alpha = 0.5$, $df_1 = 6 - 1 = 5$, $df_2 = 1{,}000 - 6 = 994$) = 2.21

Reject H_0: all beta's are simultaneously set equal to 0.



STANDARD NORMAL TABLE (Z)

Entries in the table give the area under the curve between the mean and z standard deviations above the mean. For example, for $z = 1.25$ the area under the curve between the mean (0) and z is 0.3944.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0190	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2969	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3513	0.3554	0.3577	0.3529	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998

$t_c = 1.65$

(e). first step:

$$\underbrace{\text{testing if fireplace adds value}}_{\text{is } \beta_6 \text{ significant \& positive}} \equiv \quad H_0: \beta_6 \leq 0 \quad \text{vs} \quad \beta_6 > 0$$

(if I reject the test it means it adds value)

Refresher

When you are doing one-tailed test you should always ask which test you're doing.

		Rule
right-tailed :	$H_0: \mu \leq \mu_0$ $H_1: \mu > \mu_0$	Reject if $t > t_\alpha$
left-tailed :	$H_0: \mu \geq \mu_0$ $\mu < \mu_0$	Reject if $t < -t_\alpha$

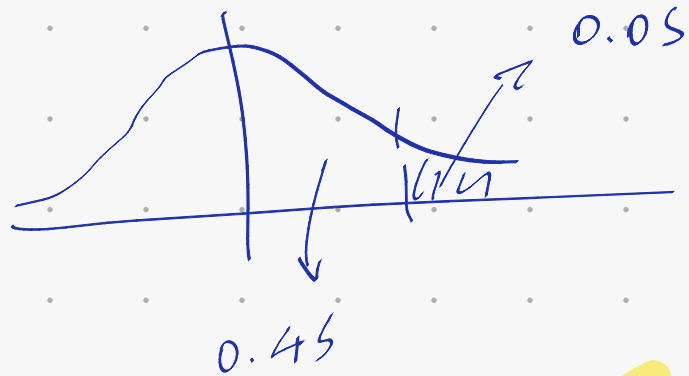
Second step:

Right-tailed test bc we're testing if parameter is positive

one tail test $H_0: \beta_6 \leq 0$ vs otherwise

$$t = \frac{\text{estimate of } \beta_2 - \text{null value}}{\text{standard error of estimate of } \beta_2} = 1.43$$

for finding critical value



$t_c = 1.645$.

The rule for right-tailed test is Reject if $t > t_\alpha$

Therefore as $1.43 \not> 1.645$ we fail to reject $\beta_6 \leq 0$.

otherwise if we have been testing $H_0: \beta_f \geq 0$ vs otherwise

we would have the rule would have been

reject if $t < -t_\alpha$.

where t_α is the number that leaves us with $\underline{1-\alpha}$

area under the bell curve.

imagine in this case we were testing

$H_0: \beta_6 \geq 0$ versus H_1: otherwise

<u>the Rule</u> (left-tail): Reject if t-stat $< -t_\alpha$

we know t-stat $= 1.43$

we know $t_{0.05} = 1.645$

therefore we fail to reject the null that $\beta_6 \geq 0$.

Problem 2 (20 points):
Consider a simple regression in which the dependent variable MIM = mean income of males who are 18 years of age or older, in thousands of dollars. The explanatory variable PMHS = percent of males 18 or older who are high school graduates. The data consists of 51 observations on the 50 states and District of Columbia. Thus, MIM and PMHS are "state averages". The estimated regression, along with standard errors and t-statistics, is

$$\widehat{MIM} = (a) + 0.180PMHS$$
$$(se) \quad (2.174) \quad (b)$$
$$(t) \quad (1.257) \quad (5.754)$$

(a) What is the estimated equation intercept? Show your calculation. (5 points)

$b_1 = t*se(b_1) = 1.257*(2.174) = 2.73$

(b) What is the standard error of the estimated slope? (5 points)

$se(b_2) = b_2/t = 0.180/5.754 = 0.0313.$

(c) State the economic interpretation of the estimated slope. Is the sign of the coefficient what you would expect from economic theory? (5 points)

The estimated slope $b_2 = 0.18$ indicates that a 1% increase in males 18 and older who are high school graduates, increase average income of those males by $180. The positive sign is as expected. more education should lead to higher salaries.

(d) Test the hypothesis that the slope of the relationship is 0.2 against the alternative that it is not. State in words the meaning of the null hypothesis in the context of this problem. (5 points)

$H_0: \beta_2 = 0.2$ vs. $H_A: \beta_2 \neq 0.2$

$$t = \frac{0.180 - 0.2}{0.0313} = -0.639 \qquad t_c = 2.009 \text{ or } -t_c = -2.009$$

Do not reject $H_0: \beta_2 = 0.02$

The null hypothesis suggests that a 1% increase in males 18 or older, who are high school graduates, leads to an average income for those males of $200.

(Non-rejection of H_0 means that this claim is compatible with the sample of the data.)

Problem 3 (16 points):
State with brief reasons whether the following statements are True, False, or Uncertain.

(a) The adjusted and unadjusted R^2's are identical when the unadjusted R^2 is equal to 1. (4 points)

True. Easy to solve which using the adjusted R^2 equation.

(b) The way to determine whether a group of explanatory variables exerts significant influence on the dependent variable is to see if any of the explanatory variables have a significant t statistic; if not they are statistically insignificant as a group. (4 points)

False. Use the F test.

(c) To compute the critical *t*, we need know only the degrees of freedom. (4 points)

False. We also need to know the level of significance.

(d) Find the critical t value for the following situation: degrees of freedom = 12, level significance = 5%, and you are conducting a two-tailed test. (4 points)

2.179

Adjusted Rsq : $\bar{R}^2 = 1 - \left(\frac{SSR / (n-k-1)}{TSS / (n-1)} \right)$

unadjusted Rsq : $R^2 = 1 - \frac{SSR}{TSS}$

when $\bar{R}^2 = 1 \implies \frac{SSR / (n-k-1)}{TSS / (n-1)} = 0 \iff SSR = 0$

$\implies R^2 = 1$ ✓

from Discussion 3.

$n = 1500$

$k = 1$

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.792876888
R Square	0.62865376
Adjusted R Square	0.628405865
Standard Error	38556.80727
Observations	1500

ANOVA

	df	SS	MS	F	Significance F
Regression	1	3.77004E+12	3.77004E+12	2535.971095	0
Residual	1498	2.22697E+12	1486627387		
Total	1499	5.99701E+12			

→ R

→ SSR

→ TSS

$$R^2 = 1 - \frac{2.22697 * 10^{12}}{5.997 * 10^{12}} = 1 - \frac{2.22697}{5.9970} = 0.628$$